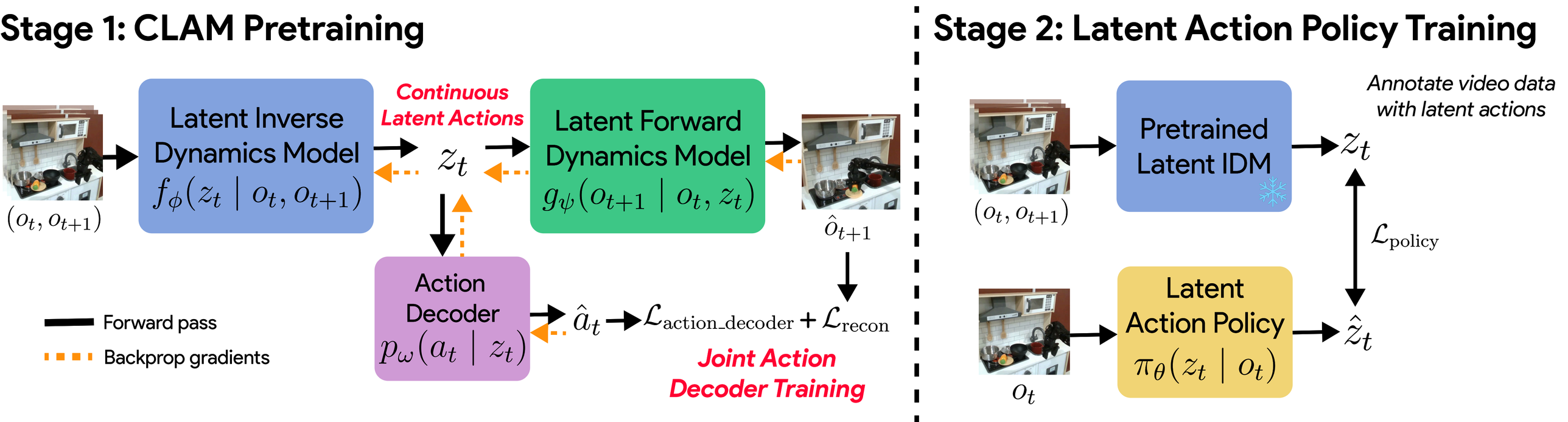
Stage 1: CLAM Pretraining
(o_t, o_{t+1})
Latent Inverse Dynamics Model
$f_\phi(z_t \mid o_t, o_{t+1})$
Continuous Latent Actions
z_t
Latent Forward Dynamics Model
$g_\psi(o_{t+1} \mid o_t, z_t)$
$\hat{o}_{t+1}$
Action Decoder
$p_\omega(a_t \mid z_t)$
$\hat{a}_t$
$\mathcal{L}_{\text{action_decoder}} + \mathcal{L}_{\text{recon}}$
Joint Action Decoder Training
Forward pass
Backprop gradients
Stage 2: Latent Action Policy Training
Annotate video data with latent actions
(o_t, o_{t+1})
Pretrained Latent IDM
z_t
$\mathcal{L}_{\text{policy}}$
o_t
Latent Action Policy
$\pi_\theta(z_t \mid o_t)$
$\hat{z}_t$